Evaluation of Force Localization in a FBG-Based Forearm Sensor by Feature Extraction and Clustering

Carlos Arturo Narvaéz Delgado[1], Luis J. Arciniegas-Mayag[1], Patrick Silva[1], Carlos A. Cifuentes[2], *Senior Member, IEEE*, Moisés R. N. Ribeiro[1], Camilo A. R. Díaz[1], *Member, IEEE*

2025 SBMO/IEEE MTT-S International Microwave and Optoelectronics Conference (IMOC) | 979-8-3503-9275-3/25/$31.00 ©2025 IEEE | DOI: 10.1109/IMOC65414.2025.11365617

***Abstract*—Evaluation of the physical Human-Robot Interface (pHRI) on exoskeletons remains a challenge due to anatomical variability, complexity of human motion, and limitations of conventional sensors. In this work a 3D printed structure that simulates a forearm and employs fiber Bragg grating (FBG) sensors, capable of detecting and classifying the location of forces applied at multiple pressure points, is presented. Classification is based on pairs and triplets of statistical features, combined with clustering analysis using the k-means algorithm. The cluster purity metric on the best feature pair had a value of 0.933, while the best two feature triplets have a purity of 1.000, evidencing a high correspondence between the generated clusters and the actual force locations. These results support the usefulness of the sensor for improving pHRI characterization in wearable devices such as exoskeletons.**

***Index Terms*—Sensor, Fiber Bragg Grating (FBG), Features, Clustering, Purity, Silhouette Coefficient, Exoskeleton, Force localization.**

I. Introduction

Exoskeletons are defined as wearable devices that enhance a subject's movement and motor skills by applying forces to the limbs through physical interfaces. In addition, these devices have experienced exponential growth over the past decade [1]. This rapid development has led to their presence in multiple healthcare, industrial, and consumer contexts, leading to the development of numerous products and an increasing number of prototypes preparing for market uptake [2]. However, exoskeletons must physically interact with the wearer in a way that is both comfortable and effective, a goal that is challenging to accomplish due to the variability of human anatomy, both between individuals and within a single user over time. As the global exoskeleton market grows, understanding the impact of the physical Human-Robot Interface (pHRI) on user comfort and performance becomes increasingly crucial [3].

The pHRI is fundamental in the development of exoskeletons, as it influences the comfort, safety and efficiency of wearable devices. The quality of the pHRI depends on factors such as joint alignment, force distribution and how the wearable devices adapt to the user's anthropometry [4]. Recent studies have shown that uncertainties in the physical-mechanical interface of wearable robotic devices can generate unintended interaction forces that reduce the effectiveness of assistance and may cause discomfort or injury to the user [5], [6].

Directly and qualitatively evaluating the pHRI can be difficult given factors such as sensor placement, human variability, metrics, and adopted protocols hinder systematic evaluation. Moreover, ethical concerns often arise when involving human subjects in early development stages, where rapid development-verification iterations are required. While modeling pHRI using spring-damper or Digital Human Models (DHM) can reduce human involvement, these models still need validation with real-life measurements, which remains challenging due to the complexity and variability of the human body. Additionally, the relative motion of the device over the body introduces further complications [2]. Therefore, the measurement of forces and torques is essential to guarantee the safety and proper functioning of robotic systems in the pHRI. There are various types of sensors used for this purpose, including resistive, piezoelectric, strain gauge-based, capacitive and optical sensors [7], [8].

In recent years, interest in fiber optic-based sensors has gained attention, especially for robotics and biomedical engineering applications, being an alternative to traditional sensors for the mechanical assessment [9]. Fiber optic-based sensors have an ease of integration into multiple types of structures. Particularly, fiber Bragg grating (FBG) sensors have a high sensitivity and ability to detect temperature variation. For instance, the work of Golmohammadi *et al.* [10] use FBG sensors for structural monitoring, and employing clustering algorithms such as Density-Based Spatial Clustering of Applications with Noise (DBSCAN) to handle large volumes of data, allowing the detection of anomalies in the stress patterns of structures. Similarly, Lin *et al.* [11] integrate Principal Component Analysis (PCA) for dimensionality reduction in data generated by FBG sensors, optimizing data classification and improving system efficiency. Likewise, D'Abbraccio *et al.* [9] present the development of an FBG-based artificial skin for applications in collaborative robotics. The skin offers advantages such as high spatial resolution, integration flexibility, and

This work is partially supported by FAPES (EXTRATO ATA DA 6ª REUNIAO CCAF/2022 – "CPID 2.2", FAPES/ CNPq/Decit – PPSUS 17/2024), CNPq (306323/2024-9, 404111/2023-8, and 402739/2024-8), and IEEE RAS SPARX.

[1] Telecommunications Laboratory (LabTel), Electrical Engineering Department, Federal University of Espírito Santo (UFES), Vitória, Brazil. carlos.delgado@edu.ufes.br, luis.mayag@edu.ufes.br, patrick.silva.48@edu.ufes.br, moises.ribeiro@ufes.br, camilo.diaz@ufes.br

[2] Bristol Robotics Laboratory, University of the West of England, Bristol, United Kingdom. Carlos.Cifuentes@uwe.ac.uk

multiplexing to reduce wiring.

Advanced FBG sensor configurations have been developed to minimize force-torque coupling effects, improving the quality and accuracy of FBG sensor measurements [12]. In this context, this work present a structure intrumented with FBG sensors, which is intended to measure the pHRI from points of contact with respect to the points of attachment of exoskeletons in the forearm. The proposed structure simulates the part of the forearm where the exoskeleton should be fixed. For this, the structure is developed taking into account anthropometric considerations. Therefore, this forearm replication will serve as a testing platform to evaluate the physical interaction forces generated by upper limb exoskeletons such as presented by Arciniegas-Mayag *et al.* [13]. This wearable device assists elbow flexion in users with hemiparesis by estimating torque on the affected limb, based on the kinematics of the unaffected limb.

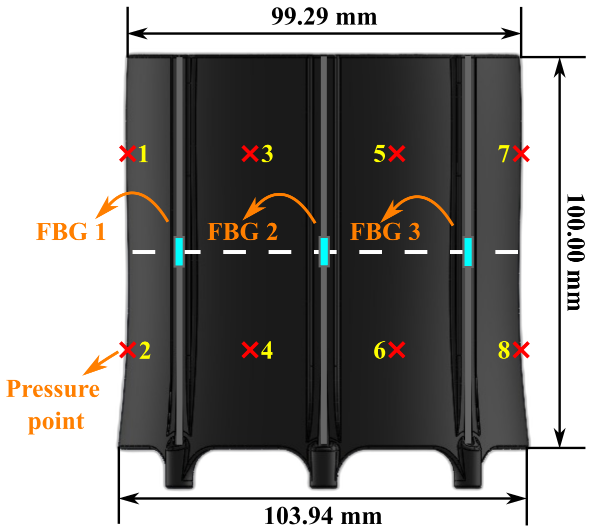

Fig. 1. Distribution of pressure points and FBG sensors in the PLA structure.

II. Methodology

The methodology implemented in this study comprises the design and fabrication of a structure with FBG sensors, as well as the experimental setup, data acquisition and response analysis. Throughout this section, the stages of construction of the structure, its physical layout, the test protocol applied and the methods used for the processing and evaluation of the information obtained are detailed.

A. Structure Design

The structure was designed in SolidWorks (Massachusetts, USA) and printed with polylactide (PLA) material. The dimensions were based on anthropometric measurements from U.S. Army databases with the 50th percentile [14].

B. FBG Sensors

Three FBG sensors were fabricated using a UV laser (CNI DPS-266, China) operating at a wavelength of 266 nm at a repetition rate of 100 Hz and a phase mask (Ibsen Photonics, Denmark). The Bragg reflectors can modify the rate of the backscattered light by generating a wavelength shift when a perturbation is applied to the FBG. This is important because the optical fibers, along with the FBG sensors, must be subjected to strained in order to detect perturbations. When attached to the structure shown in Figure 1, the sensors enable the measurement of structural deformations resulting from applied forces. The wavelengths of the FBG sensors are 1548.4 nm, 1548.2 nm, 1562.5 nm.

C. Sensor Configuration and Experimental Setup

After placing the FBG sensors on the structure, the structure was divided into eight pressure points, with the purpose of applying forces in each section. This segmentation enables the identification of the force application point. Figure 1 shows the positioning of the FBG sensors and the location of the pressure points.

A setup has been developed to use a micrometer screw (263MXL, Starrett, USA) and a strain gauge (LCM201, Omega, USA) to apply force at the established points, as shown in Figure 2. The micrometer screw is used to apply a controlled force on the surface of the structure. The magnitude of the applied force corresponds to the normal force respect to the surface of the structure. In addition, the developed configuration is designed to facilitate the rotation and translation of the structure in order to apply the force at different points.

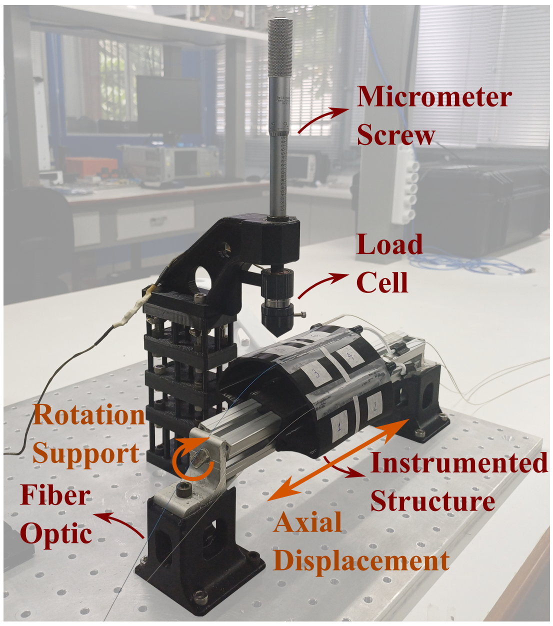

Fig. 2. Instrumented structure and experimental setup.

D. Data Acquisition and Processing

Data collection from the FBG sensors was performed using the HYPERION si255 (Luna's, USA) interrogator with a sampling rate of 1000 sps. A mean-based interpolation was implemented to fill in any missing data, generated by low amplitude in the backscattered spectrum, specially in FBG 1.

In addition, outliers were removed using the interquartile range, and a 5 Hz low-pass filter was applied to reduce high frequency noise.

After pre-processing stage, the following features were extracted: Mean, Standard Deviation (SD), Maximum (Max), Minimum (Min), Energy (En), Waveform Length (WL), Skewness (Skew) and Kurtosis (Kurt). For the analysis of these features, a Z-score normalization was applied to ensure all features had a mean of 0 and a standard deviation of 1.

To assess the discriminatory power of the extracted features, all possible combinations of two and three variables were analyzed using validation metrics based on clustering techniques. Each subset of features was used to cluster the samples using the k-means algorithm. The quality of the resulting partitions was assessed using two criteria. First, Cluster Purity was calculated as an external evaluation metric, in order to quantify the agreement between the predicted clusters and the true class labels; this metric takes values between 0 and 1, where higher values indicate a better match [15]. Second, the Silhouette Coefficient was used as an internal metric, which measures intracluster cohesion and intercluster separation; its values range from -1 to 1, with higher values being indicative of more compact and well-defined clusters [16].

E. Response Analysis and Sensor Behavior

In order to know the dynamic behavior of the FBG sensors when applying force on the developed structure, the following test was performed:

- A force was applied in each of the eight pressure points.
- The applied force was 16.73 ± 0.61 N.
- 15 tests were performed per pressure point to evaluate the stability and response of the sensors. Each test consisted of 3 stages:
  - 30 seconds without applying force.
  - 30 seconds applying force.
  - 30 seconds without applying force.

III. Results

The developed structure deforms depending on the location of the applied forces. The distribution of the FBG sensors allows quantifying the deformations. Figure 3 shows that the signals generated by the different pressure points present a differentiable behavior that allows their classification.

In order to identify the combinations of features that best classify the location of the applied force at the eight pressure points of the structure, an analysis based on clustering and classification metrics was carried out. The results are organized into two subsections according to the number of combined features: pairs and triplets.

A. Pairs of Features

This section presents the combinations of two features that offer the highest inter-class separability according to the Cluster Purity and Silhouette Coefficient as shown in Table I

As some combinations offer similar overall performance, cluster purity is prioritized as the main ranking criterion due to its supervised nature based on real labels. Consequently, Figure 4 presents the scatter plot corresponding to the best ranked feature pair.

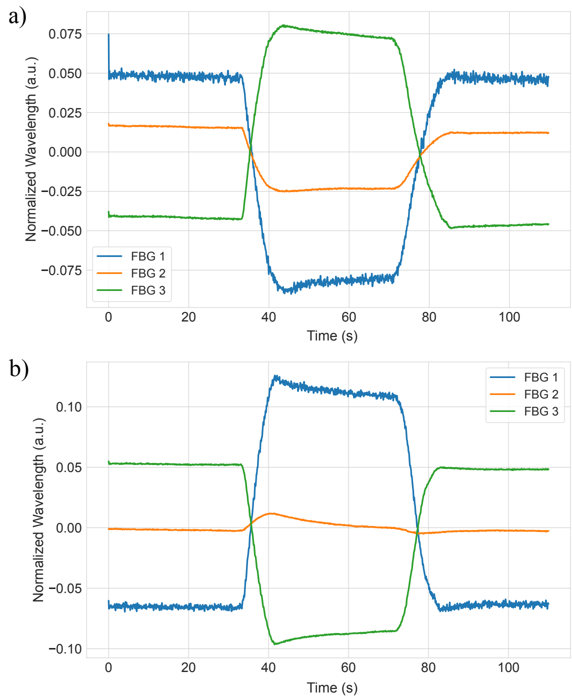

Fig. 3. a) Plot of the mean of the filtered data centered on the mean, for pressure point 4. b) Plot of the mean of the filtered data centered on the mean, for pressure point 6.

TABLE I
Top 5 Feature Pairs Ranked by Cluster Purity (Primary Criterion) and Silhouette Coefficient (Secondary Criterion)

Features		Cluster Purity	Silhouette Coefficient
Feature 1	Feature 2		
FBG 1 (SD)	FBG 2 (SD)	0.933	0.585
FBG 1 (SD)	FBG 3 (Skew)	0.875	0.689
FBG 1 (SD)	FBG 1 (Skew)	0.875	0.577
FBG 2 (SD)	FBG 3 (SD)	0.858	0.490
FBG 1 (SD)	FBG 3 (SD)	0.833	0.551

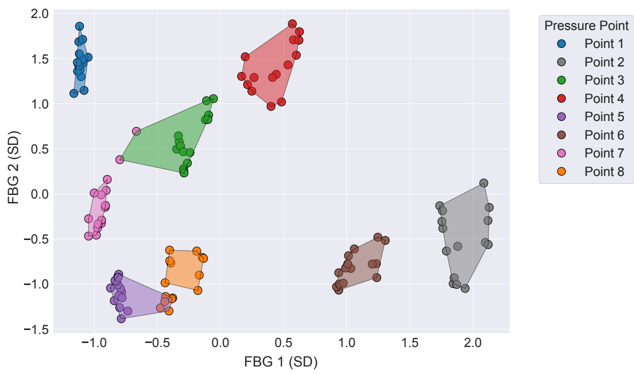

Fig. 4. Scatter plot and clustering (using the k-means algorithm with $k = 8$) based on the standard deviation of FBG 1 and FBG 2. The clusters are represented by shaded regions bounded by convex hulls and colored according to the dominant pressure point in each cluster.

B. Triplets of Features

This section presents the results for combinations of three features. Table II summarizes the five best performing combinations according to the same parameters. Figure 5 shows the scatter plot of the best triplet of features identified during the analysis.

TABLE II
TOP 5 FEATURE TRIPLETS RANKED BY CLUSTER PURITY (PRIMARY CRITERION) AND SILHOUETTE COEFFICIENT (SECONDARY CRITERION)

Features			Cluster Purity	Silhouette Coefficient
Feature 1	Feature 2	Feature 3		
FBG 1 (SD)	FBG 3 (SD)	FBG 3 (Skew)	1.000	0.751
FBG 1 (SD)	FBG 2 (SD)	FBG 3 (Skew)	1.000	0.648
FBG 2 (SD)	FBG 3 (SD)	FBG 3 (Skew)	0.992	0.615
FBG 1 (SD)	FBG 1 (Skew)	FBG 3 (SD)	0.983	0.681
FBG 1 (SD)	FBG 2 (SD)	FBG 3 (SD)	0.983	0.672

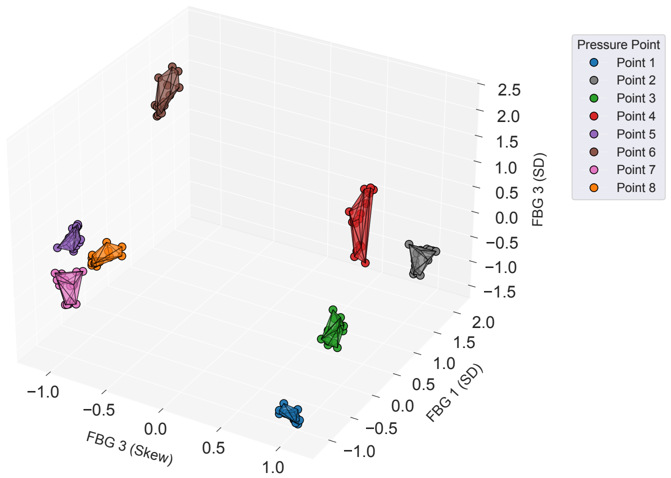

Fig. 5. Scatter plot and clustering (using the k-means algorithm with $k = 8$) based on the standard deviation of FBGs 1 and 3, and the skewness of FBG 3. The clusters are represented by shaded 3D regions bounded by polyhedral convex hulls and are colored according to the dominant pressure point in each cluster.

IV. DISCUSSIONS

The results obtained indicate the feasibility of using FBG sensors to differentiate the location of the forces applied at eight different pressure points on a structure that simulates the forearm.

The most representative feature combinations have a high Cluster Purity indicating that the groups generated by the clustering algorithm largely match the actual pressure point labels. This classification capability is similar to the approaches of Golmohammadi and Lin [10], [11], who use FBG to classify stress data in structures, although, in this study, high clustering purity is achieved without the need for methods such as dimensionality reduction using PCA.

Table II shows that the combination of characteristics of the first two triplets present a maximum purity. However, they differ greatly from the metric of the Silhouette Coefficient. Therefore, the data of the first triplet of characteristics presents a better grouping internally and a better separation between classes. Figure 4 shows how two of the clusters generated by the k-means algorithm have data referring to another pressure point, this error in classification is directly related to the Cluster Purity as seen in Table I. In Golmohammadi's work [10], the accuracy in the classification depends on the selected characteristics, such as SD and Skew. In this research, these characteristics allowed a better separation between classes. FBG 2 provides variability in its data as seen in Tables I and II. The characteristics in which FBG 2 is present show a reduction of the Silhouette Coefficient compared to the characteristics in which FBGs 1 and 3 are present exclusively. Figure 5 shows that the k-means algorithm achieved cluster formation with a purity of value 1, indicating a complete correspondence between the identified clusters and the actual labels. The difference in the Silhouette Coefficient is reflected when comparing Figures 4 and 5 since the clusters of Figure 5 are more compact and distant. In addition, Figure 5 shows that the Skew characteristic provided by FBG 3 allows to identify whether the force was applied to the right or left of the structure because the points below -0.5 are the points corresponding to the right side of the structure.

The above reinforces the hypothesis that a suitable configuration of FBG sensors, together with an appropriate statistical processing, allows to accurately characterize the location of the pressure points on a structure simulating the forearm. While the work of D'Abbraccio *et al.* [9] uses sixteen FBG sensors for quantification of the force applied on a large area of artificial sensory skin on a humanoid robotic forearm. The presented results show that it is possible to use few FBG sensors for an optimal localization of the pressure points. Finally, the results obtained open possibilities for the use of this type of sensors in the design and validation phases of exoskeletons, allowing repeatable tests without resorting directly to human users.

V. CONCLUSIONS AND FUTURE WORK

In this work a structure intrumented with FBG sensors was developed to evaluate the pHRI in the context of exoskeletons, specifically focused on the forearm. The design of the structure, based on anthropometric measurements, made it possible to simulate a representative setting and contact environment on which controlled forces were applied at eight pressure points.

The results obtained show that it is possible to adequately detect and classify the location of the forces applied on the designed structure, due to the strategic distribution of the FBG sensors and the analysis of the statistical characteristics extracted from the signals Combinations of characteristics such as standard deviation and skewness of the FBG sensors signals demonstrated a high discriminative capacity validated by metrics such as the Silhouette coefficient and Cluster Purity.

The high performance in these metrics indicates not only a good separation between the pressure classes, but also a strong correspondence between the clusters obtained and the actual locations, supporting the use of this type of structure as a tool to evaluate and optimize the design of physical interfaces in portable devices.

Since the results of this study show that the identification of the applied force at the established pressure points may require only two FBG sensors, it is left as future work to quantify the magnitude of the applied force at each pressure point.

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